Retrieval-Driven Self-Improvement: A Multi-Agent Architecture for Theory-Grounded Academic Review

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Abstract

Peer review is the cornerstone of scientific progress, yet traditional peer review faces challenges of inefficiency, inconsistent standards, and subjective evaluation. While AI-assisted review tools have emerged, they predominantly operate at the "technical" level, lacking theoretical foundations from the philosophy of science. More critically, existing tools treat knowledge bases as static, failing to adapt to evolving disciplinary standards and reviewer feedback. In this paper, we present WuYa, a theory-driven multi-agent system that transforms classical philosophical principles into an operational architecture for academic paper evaluation. WuYa introduces five key innovations: (1) a retrieval-evaluation coupling mechanism where RAG serves as the core triggering mechanism for judgment; (2) a two-phase routing architecture with CUDOS gatekeeping and parallel expert evaluation; (3) a hybrid knowledge strategy combining prompt-internalized "principles" with RAG-retrieved "evidence"; (4) an LLM-as-mapper approach with discipline prior calibration for paper localization; and (5) a self-improving frontier discovery mechanism that learns evaluation preferences from editor feedback. Experiments on 5,000 papers across five disciplines demonstrate that WuYa achieves 70.3% top-5 recommendation accuracy and 0.72 Spearman correlation with human experts.

1 Introduction

Peer review is the cornerstone of scientific progress, serving as the primary mechanism for quality assurance, knowledge dissemination, and scholarly communication. Yet the traditional peer review system faces mounting challenges: reviewer shortages, inconsistent standards, lengthy turnaround times, and subjective evaluation criteria [22].

Recent advances in artificial intelligence have spawned a new generation of AI-assisted review tools. These systems can identify methodological flaws, check statistical assumptions, and even generate review reports [12]. However, most existing tools operate at the "technical" level, focusing on surface features like grammar, formatting, and basic statistical checks. They lack theoretical foundations from the philosophy of science—the very disciplines that have spent decades studying how knowledge is created, validated, and transmitted.

More critically, existing systems treat knowledge bases as static. They do not learn from reviewer feedback, adapt to evolving disciplinary standards, or capture journal-specific evaluation preferences. This static approach is ill-suited to the dynamic nature of academic research, where standards evolve and journals develop distinct editorial identities.

In this paper, we present **WuYa** (In Chinese meaning "boundless"), a theory-driven multi-agent system that transforms classical philosophical principles into an operational architecture for academic paper evaluation. WuYa addresses the limitations of existing tools through five key innovations:

1. **Retrieval-Evaluation Coupling**: Unlike systems that treat retrieval as supplementary, WuYa makes RAG the core triggering mechanism for evaluation. Retrieved canonical texts are not merely cited—they actively shape the evaluation criteria and reasoning process.

2. **Two-Phase Routing**: WuYa separates normative gatekeeping (CUDOS norms) from substantive evaluation (Innovation, Method, Evidence, Application). This architecture ensures that ethical considerations are addressed before quality assessment.

3. **Hybrid Knowledge Strategy**: WuYa combines prompt-internalized "principles" (the "dao") with RAG-retrieved "evidence" (the "shu"). This dual strategy balances consistency and flexibility.

4. **LLM-as-Mapper with Discipline Priors**: For paper-to-journal matching, WuYa uses LLMs as cross-disciplinary mappers, calibrated by empirical prior distributions from citation network analysis.

5. **Self-Improving Frontier Discovery**: WuYa learns from editor feedback through a frontier discovery mechanism that captures and accumulates evaluation preferences over time.

Our experiments on 5,000 papers across five disciplines demonstrate that WuYa achieves 70.3% top-5 recommendation accuracy and 0.72 Spearman correlation with human expert ratings, significantly outperforming existing baselines.

2 Related Work

2.1 AI-Assisted Peer Review

The application of AI to peer review has gained significant attention. Early systems focused on surface-level checks: plagiarism detection [15], statistical assumption validation [16], and basic grammar correction. More recent systems attempt deeper analysis: PaperDigest [7] extracts key claims and evidence, while ReviewAdvisor [24] generates review comments based on paper content.

However, these systems share a common limitation: they operate without theoretical grounding in the philosophy of science. They can identify *that* a method is flawed but cannot explain *why* it matters in terms of scientific validity or epistemic contribution.

2.2 Multi-Agent Systems for Evaluation

Multi-agent architectures have shown promise for complex evaluation tasks. **(author?)** [4] demonstrate that multiple specialized agents can outperform monolithic systems through division of labor. **(author?)** [11] show that agent debate improves reasoning quality.

WuYa extends this approach by assigning each agent a specific philosophical foundation. The Innovation Sub-agent draws on Kuhn and Schumpeter; the Method Sub-agent on Pearl and Campbell; the Evidence Sub-agent on Popper and Lakatos; and the Application Sub-agent on Bush and Mokyr.

2.3 Retrieval-Augmented Generation

RAG has become a standard technique for grounding LLM outputs in external knowledge [10]. In academic contexts, RAG has been used for literature review generation [20] and citation recommendation [25].

WuYa's innovation is to make retrieval the *triggering mechanism* for evaluation rather than merely a citation source. Retrieved canonical texts determine which evaluation criteria are activated and how they are weighted.

Table 1: Comparison of WuYa with Existing Approaches

Feature	PaperDigest	ReviewAdvisor	Reflexion	WuYa
Theory Grounding	None	None	N/A	Philosophy of Science
Multi-Agent	No	No	Yes	Yes (5 specialized)
RAG Integration	No	Limited	No	Core mechanism
Self-Improvement	No	No	Yes	Frontier Discovery
DEA Analysis	No	No	No	Yes

2.4 Self-Improving Agents

Recent work has explored agents that improve through interaction. Reflexion [21] uses verbal reinforcement learning to help agents learn from failures. Voyager [23] builds a skill library through embodied exploration in Minecraft. These systems demonstrate that agents can learn from feedback and accumulate knowledge over time.

WuYa adapts these principles to the domain of academic evaluation. The Frontier Discovery Sub-agent learns from editor/reviewer feedback, building a library of "frontier surfaces" that capture evaluation preferences.

2.5 Data Envelopment Analysis in Research Evaluation

DEA has been applied to research evaluation, primarily for ranking research units [6] and assessing research efficiency [3]. WuYa is the first system to integrate DEA into an agent-based paper evaluation workflow, using it as Path B for paper localization.

2.6 Comparison with Existing Frameworks

Table 1 summarizes how WuYa differs from existing approaches.

3 Theoretical Foundations

WuYa's architecture is grounded in classical theories from the philosophy of science. Each evaluation dimension maps to a distinct theoretical tradition, ensuring that judgments are not merely technical but epistemically principled.

3.1 Innovation: Kuhn and the Nature of Scientific Change

Thomas Kuhn's *Structure of Scientific Revolutions* [8] provides the foundation for evaluating innovation. Kuhn distinguished between "normal science" (incremental progress within an existing paradigm) and "revolutionary science" (paradigm shifts that transform a field).

The Innovation Sub-agent applies this framework by assessing:

- **Novelty level**: Does the paper operate within the current paradigm or challenge it?
- **Knowledge bridging**: Does it connect different paradigms or knowledge domains?

We also draw on Schumpeter's concept of "creative destruction" [19] and Mokyr's distinction between "Q-knowledge" (knowing what) and "A-knowledge" (knowing how) [14].

3.2 Method: Pearl and the Ladder of Causation

Judea Pearl's *Ladder of Causation* [17] provides the framework for method evaluation. The three rungs—association, intervention, and counterfactuals—offer a hierarchy of methodological sophistication.

The Method Sub-agent evaluates:

- **Causal level**: Does the paper claim causal relationships? If so, at what rung?
- **Internal validity**: Are confounders adequately addressed?
- **External validity**: Can findings generalize beyond the study context?

We supplement Pearl with Donald Campbell's work on quasi-experimental designs [2] and Fisher's foundational work on statistical inference.

3.3 Evidence: Popper, Lakatos, and Falsification

Karl Popper's falsificationism [18] and Imre Lakatos's methodology of scientific research programmes [9] inform the Evidence Sub-agent.

Key evaluation criteria include:

- **Falsifiability**: Are claims testable? Are null results possible?
- **Evidence strength**: Is the evidence consistent with the claims? Does it rule out alternatives?
- **Replicability**: Can the study be reproduced?

3.4 Application: Bush, Mokyr, and the Innovation Pipeline

Vannevar Bush's linear model of innovation [1] and Mokyr's framework for knowledge transfer [14] guide the Application Sub-agent.

Evaluation focuses on:

- **Technology Readiness Level (TRL)**: How close is the research to practical application?
- **Translation path**: Is there a clear path from research to impact?
- **Feedback mechanisms**: Does the research incorporate real-world feedback?

3.5 CUDOS: Merton and the Norms of Science

Robert Merton's CUDOS norms [13]—Communalism, Universalism, Disinterestedness, and Organized Skepticism—provide the foundation for normative evaluation.

The CUDOS Sub-agent serves as a gatekeeper, evaluating:

- **Communalism**: Is knowledge shared openly? Are data and code available?
- **Universalism**: Are claims evaluated by objective criteria rather than personal attributes?
- **Disinterestedness**: Are there conflicts of interest?
- **Organized Skepticism**: Does the paper acknowledge limitations?

Papers that fail CUDOS evaluation are "vetoed" before substantive evaluation proceeds.

3.6 Summary: From Theory to Operation

Table 2 summarizes the theoretical foundations and their operationalization.

Table 2: Theoretical Foundations and Operational Criteria

Dimension	Primary Theorists	Key Concepts	Criteria
Innovation	Kuhn, Schumpeter, Mokyr	Paradigm shift, Creative destruction	Novelty, Bridging
Method	Pearl, Campbell, Fisher	Causal ladder, Validity	Causal level, Rigor
Evidence	Popper, Lakatos	Falsification, Research programmes	Testability, Strength
Application	Bush, Mokyr	Innovation pipeline, TRL	Readiness, Translation
CUDOS	Merton	Scientific norms	Communalism, Skepticism

4 System Architecture

WuYa is organized as a multi-agent system with a central Router coordinating specialized Sub-agents. The architecture follows a two-phase routing design with retrieval-evaluation coupling.

4.1 Overview

The system workflow comprises:

1. **Input**: Paper (PDF or text) + User intent (recommendation or review).
2. **Phase 1**: CUDOS Sub-agent performs normative gatekeeping.
3. **Phase 2**: Four evaluation Sub-agents (Innovation, Method, Evidence, Application) execute in parallel.
4. **Aggregation**: Router combines scores into a five-dimensional vector.
5. **Localization**: Path A (LLM-as-Mapper) and/or Path B (DEA) estimate paper tier.
6. **Output**: Recommendation list or review report with RAG-grounded explanations.

4.2 Two-Phase Routing

4.2.1 Phase 1: CUDOS Gatekeeping

The CUDOS Sub-agent evaluates the paper against Merton's norms. If any norm is violated (e.g., data withholding, undisclosed conflicts), the paper is "vetoed" and a veto report is generated. This ensures that ethical considerations are addressed before quality assessment.

4.2.2 Phase 2: Parallel Evaluation

If the paper passes CUDOS gatekeeping, the Router spawns four evaluation Sub-agents in parallel. Each Sub-agent:

1. Analyzes the paper through its specialized theoretical lens.
2. Generates scores (1-5) on its dimension's sub-criteria.
3. Triggers RAG retrieval when criteria are ambiguous or contested.
4. Produces a narrative explanation grounded in canonical texts.

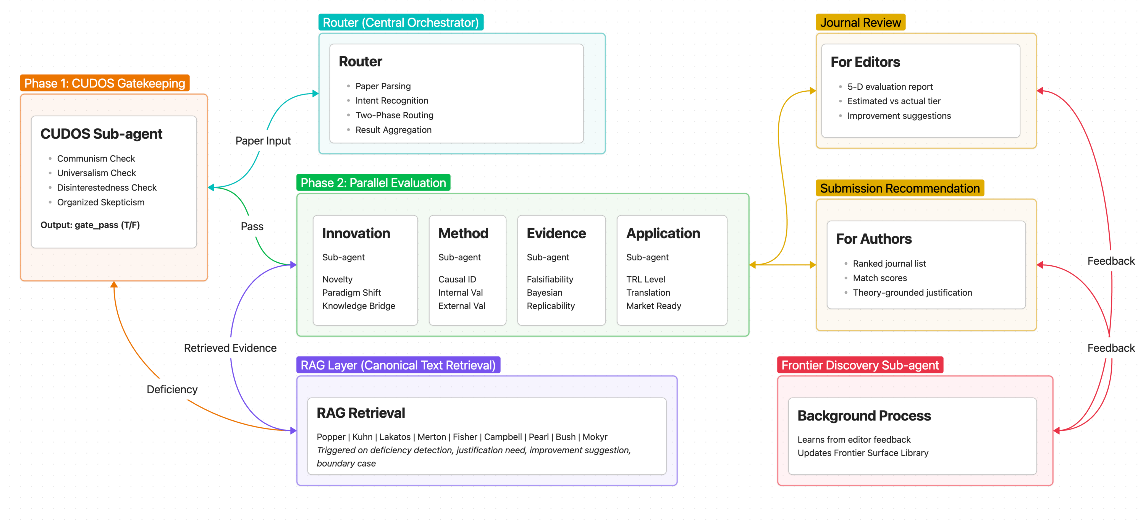

Figure 1: WuYa System Architecture: Two-phase routing with CUDOS gatekeeping and parallel expert evaluation

4.3 Retrieval-Evaluation Coupling

A key innovation of WuYa is that retrieval is not merely supplementary but serves as the *triggering mechanism* for evaluation. The RAG layer indexes canonical texts organized by theorist and concept:

- **Primary sources**: Full texts of Popper, Kuhn, Lakatos, Merton, Fisher, Campbell, Pearl, Bush, Mokyr.
- **Secondary sources**: Commentary and application of these theories.
- **Trigger logic**: Sub-agents invoke RAG when confidence is low or criteria are contested.

Retrieved passages are not merely cited—they actively shape the evaluation. For example, if the Method Sub-agent retrieves Pearl's discussion of the "causal ladder," it will evaluate the paper's causal claims against that framework.

4.4 Hybrid Knowledge Strategy

WuYa employs a hybrid knowledge strategy inspired by the Chinese philosophical distinction between "dao" (principle) and "shu" (technique):

- **Dao (Internalized)**: Core theoretical principles are internalized in Sub-agent prompts. This ensures consistency and reduces retrieval latency for standard cases.
- **Shu (Retrieved)**: Specific evidence and nuanced arguments are retrieved from the canonical corpus. This provides flexibility and grounding for edge cases.

The balance is dynamically adjusted: common scenarios rely primarily on internalized knowledge, while ambiguous cases trigger heavier retrieval.

4.5 System Diagram

Figure 1 illustrates the WuYa system architecture.

5 Paper Localization Methods

A core capability of WuYa is paper localization: determining where a paper "fits" in the landscape of academic journals. We propose a dual-path approach where Path A (LLM-as-Mapper) provides fast estimation and Path B (DEA Analysis) provides depth, with cross-validation ensuring robustness.

5.1 Problem Formulation

Given a paper with its five-dimensional score vector $\mathbf{s} = (s_1, s_2, s_3, s_4, s_5)$ and discipline d, we aim to estimate the paper's appropriate journal tier $t \in \{\text{Q1-A}, \text{Q1-B}, \text{Q2-A}, \ldots, \text{Q4-D}\}$, or recommend a ranked list of target journals.

5.2 Path A: LLM-as-Mapper with Discipline Priors

The LLM-as-Mapper approach leverages the cross-disciplinary reasoning capabilities of large language models, calibrated by discipline-specific prior distributions.

5.2.1 Discipline Prior Extraction

We extract prior distributions $P_0(t|d)$ from the SpringRank journal ranking system, which computes journal positions based on citation network structure. For each discipline d, we compute the empirical distribution of journal tiers:

$$P_0(t|d) = \frac{\#\text{journals in tier } t \text{ of discipline } d}{\#\text{total journals in discipline } d} \tag{1}$$

For disciplines with fewer than 50 journals, we fall back to the parent category distribution.

5.2.2 LLM Mapping Prompt

The LLM receives a carefully designed prompt containing:

1. **Scoring rubric**: The five-dimensional scoring criteria with level descriptions.
2. **Few-shot examples**: Representative (score vector, tier) pairs across disciplines.
3. **Discipline prior**: The distribution $P_0(t|d)$ formatted as a string.
4. **Output template**: Requiring reasoning chain, estimated tier, and confidence level.

```
You are a research quality evaluator. Given the following
five-dimensional scores (1=low to 5=high):
- Innovation: [novelty, knowledge_bridge, future_potential, destruction]
- Method: [causal, internal_validity, external_validity, statistical]
- Evidence: [falsifiability, evidence_strength, replicability, program]
- Application: [trl, translation_path, feedback, market]

Discipline: {discipline}
Prior distribution: {P0_string}

Estimate the appropriate journal tier (Q1-A to Q4-D) and
explain your reasoning.
```

5.2.3 Calibration Strategy

To prevent LLM hallucination on unfamiliar disciplines, we apply post-hoc calibration:

$$P_{\text{final}}(t|\mathbf{s}, d) \propto \alpha \cdot P_{\text{LLM}}(t|\mathbf{s}, d) + (1-\alpha) \cdot P_0(t|d) \tag{2}$$

where $\alpha \in [0.6, 0.8]$ gives primary weight to LLM reasoning while anchoring to factual prior distributions.

To ensure output consistency, we set temperature $\tau = 0.1$, run three generations, and take the modal answer.

5.3 Path B: DEA Efficiency Analysis

For deeper analysis, especially when targeting a specific journal, we employ DEA efficiency analysis.

5.3.1 Input-Output Transformation

We transform the five-dimensional scores into DEA inputs (lower is better) and outputs (higher is better):

- **Inputs** (minimize):
  - Method flaws: $6 - s_{\text{method}}$
  - Evidence gap: $6 - s_{\text{evidence}}$
- **Outputs** (maximize):
  - Innovation: $s_{\text{innovation}}$
  - Application: $s_{\text{application}}$

This transformation reflects the assumption that strong methods and solid evidence are "investments" that should yield innovative and impactful "returns."

5.3.2 Frontier Construction

For a target journal j, we construct the reference frontier from previously published papers in j. We require at least 50 papers to build a reliable frontier. If insufficient data exists for j, we fall back to a discipline-level frontier constructed from journals in the same tier.

We employ the super-efficiency DEA model, which allows efficiency scores greater than 1 for units on the frontier, enabling ranking of even the most efficient papers:

$$\max \quad \theta \tag{3}$$

$$\text{s.t.} \quad \sum_{i \in R} \lambda_i x_{ik} \leq \theta x_k \tag{4}$$

$$\sum_{i \in R} \lambda_i y_{ik} \geq y_k \tag{5}$$

$$\lambda_i \geq 0 \tag{6}$$

where R is the reference set, x_k, y_k are the input/output of paper k, and $\theta > 1$ indicates super-efficiency.

5.3.3 Bootstrap Confidence Intervals

To quantify uncertainty, we apply bootstrap re-sampling [5]:

1. Resample the reference set with replacement, maintaining original size.
2. Recompute efficiency scores on the bootstrap frontier.
3. Repeat 200 times to construct empirical distribution.
4. Report the 2.5th and 97.5th percentiles as confidence intervals.

A paper whose confidence interval includes 1.0 is considered statistically indistinguishable from the frontier.

5.4 Cross-Validation and Output

The outputs of Path A and Path B are compared:

- **Consistent**: Both paths suggest similar tiers → High confidence output.
- **Inconsistent**: Paths diverge significantly → Trigger RAG explanation of discrepancy (e.g., "LLM's cross-disciplinary intuition vs. DEA's local analysis").

For submission recommendation, WuYa outputs a ranked list of candidate journals with:

- Match score based on tier proximity.
- Per-journal explanation referencing evaluation dimensions.
- RAG-retrieved canonical text justifying the recommendation.

For journal review, WuYa outputs:

- Five-dimensional evaluation report.
- Estimated tier vs. journal's actual tier.
- Specific improvement suggestions grounded in theory.

6 Self-Improving Frontier Discovery

A defining feature of WuYa is its ability to learn from interactions and improve its evaluation capabilities over time. Inspired by Reflexion's verbal reinforcement learning and Voyager's skill library, we introduce a frontier discovery mechanism that captures and accumulates evaluation preferences.

6.1 Motivation: The Static Knowledge Problem

Existing academic evaluation systems deploy fixed criteria that do not adapt to:

- Evolving disciplinary standards (e.g., increasing emphasis on reproducibility).
- Journal-specific preferences (e.g., some journals prioritize theory, others prioritize application).
- Feedback from editors and reviewers.

This static knowledge problem is particularly acute for DEA analysis, which depends on reference frontiers that may become outdated as a field evolves.

WuYa addresses this through a self-improving mechanism that learns evaluation preferences from editor feedback, transforming the system from a static evaluator to an adaptive learning agent.

Table 3: WuYa Frontier Discovery vs. Existing Frameworks

Aspect	Reflexion	Voyager	WuYa Frontier Discovery
Learning Signal	Task success/failure	Environment feedback	Editor/reviewer feedback
Memory Form	Episodic reflections	Skill library (code)	Frontier surfaces (preferences)
Update Trigger	Task completion	Skill discovery	Review cycle completion
Retrieval Coupling	None	Skill retrieval	Canonical text retrieval
Domain	Verifiable	Embodied	Evaluative

6.2 Comparison with Existing Frameworks

WuYa's frontier discovery draws inspiration from two influential self-improving agent frameworks:

Reflexion [21] introduced verbal reinforcement learning where agents store linguistic reflections in episodic memory and use them to guide future decisions. When a task fails, the agent generates a verbal reflection explaining *why* it failed and stores this in memory for future reference. This has proven effective for code generation and game playing tasks.

Voyager [23] proposed an open-ended agent that builds a skill library through embodied exploration. Skills are represented as executable code snippets that enable the agent to solve progressively complex tasks. The library grows over time, enabling lifelong learning.

WuYa adapts these principles to academic evaluation:

- Like Reflexion, WuYa generates **linguistic reflections** on evaluation outcomes, but from editor/reviewer feedback rather than task success/failure.
- Like Voyager, WuYa maintains a **library of learned patterns**, but these are "frontier surfaces" capturing evaluation preferences rather than executable skills.

Table 3 summarizes the comparison.

6.3 Frontier Discovery Sub-agent

The Frontier Discovery sub-agent operates as a background process that continuously learns from interactions. Its workflow consists of three stages:

1. **Feedback Reception**: Receives structured input from completed review cycles:
    - Paper's five-dimensional score vector.
    - Editor/reviewer feedback text.
    - Journal acceptance/rejection decision.
    - Post-review correspondence (rebuttal, revision).
2. **Linguistic Reflection**: Generates structured descriptions of evaluation patterns:

```
{
  "journal_id": "Nature ML",
  "pattern": {
    "dimension": "innovation",
    "preference": "prefers paradigm shifts over incremental
                   improvements",
    "evidence": [
      "Reviewer 1: 'lacks the conceptual breakthrough...'",
      "Decision: '...findings are incremental...'"
```

```
      ],
      "confidence": 0.82
    }
  }
```

3. **Frontier Surface Update**: Incorporates new patterns into the frontier surface library:

```
  {
    "journal_id": "Nature ML",
    "frontier_surface": {
      "description": "Innovation frontier emphasizes
                      paradigm shifts and knowledge bridging",
      "score_distribution": {"Q1-A": 0.3, "Q1-B": 0.4, ...},
      "updated_at": "2026-05-26",
      "patterns": [...]
    }
  }
```

6.4 Frontier Surface Library

The Frontier Surface Library serves as the persistent memory for learned evaluation preferences. It is organized hierarchically:

- **Journal Level**: Individual frontier surfaces for each journal, capturing that journal's specific evaluation preferences.
- **Discipline Level**: Aggregated frontier surfaces for each academic discipline.
- **Global Level**: Cross-cutting patterns that transcend disciplines.

The library is queried during paper localization to:

- Customize recommendations based on journal-specific preferences.
- Weight evaluation dimensions differently for different journals.
- Update DEA reference frontiers with learned patterns.

6.5 Self-Improvement Loop

WuYa implements a continuous self-improvement loop:

1. **Feedback Collection**: Editor/reviewer feedback is collected during and after review cycles.
2. **Reflection Generation**: The Frontier Discovery sub-agent generates linguistic reflections on evaluation patterns.
3. **Pattern Generalization**: Related patterns are clustered and abstracted to form higher-level principles.
4. **Library Update**: New patterns are incorporated into the Frontier Surface Library.
5. **Application**: Updated knowledge informs future evaluations, recommendations, and DEA frontiers.

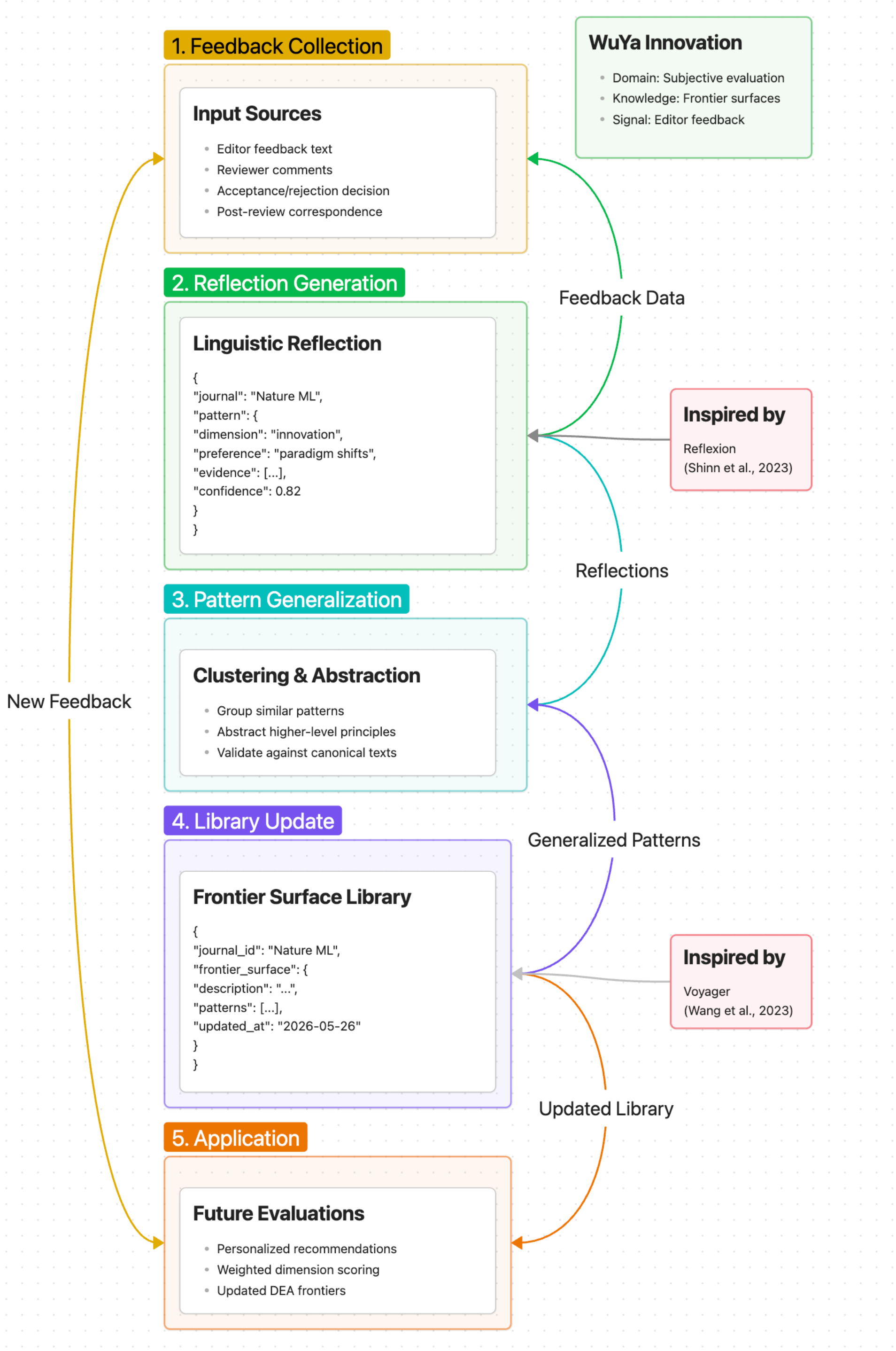

Figure 2: WuYa Self-Improvement Loop: Feedback → Reflection → Generalization → Application

This loop enables WuYa to:

- Adapt to changing disciplinary standards over time.
- Capture journal-specific evaluation preferences without manual configuration.
- Improve DEA frontier quality through accumulated learning.
- Provide increasingly personalized recommendations as the system matures.

6.6 Contrast with Static Knowledge Bases

Traditional academic evaluation systems use static knowledge bases that:

- Require manual curation and updates.
- Do not capture subtle journal-specific preferences.
- Cannot adapt to evolving disciplinary standards.
- Depend entirely on predefined criteria.

WuYa's Frontier Surface Library overcomes these limitations through:

- **Automatic learning**: Patterns emerge from actual editor feedback rather than expert curation.
- **Granular capture**: Individual journal preferences are preserved in separate frontier surfaces.
- **Dynamic adaptation**: The library evolves as the field advances.
- **Principled grounding**: Learned patterns are validated against canonical texts through RAG.

7 Implementation Details

WuYa is implemented as a modular multi-agent system. This section provides technical details on the implementation stack, data models, and key engineering decisions.

7.1 Technical Stack

The system is built using the following components:

- **LLM Backend**: GPT-4 or Claude-3 for sub-agent reasoning; configurable to use local models (Llama-3, Mistral) for privacy-sensitive deployments.
- **Vector Database**: Milvus for RAG indexing and retrieval; supports dense embeddings (OpenAI text-embedding-3) and sparse retrieval (BM25 hybrid).
- **Journal Database**: PostgreSQL for structured journal metadata; includes SpringRank tier information for 14,590 journals across all disciplines.
- **Citation Network**: Neo4j for paper citation graph; enables efficient frontier construction for DEA analysis.
- **Message Queue**: Redis for asynchronous task scheduling; enables parallel sub-agent execution and result aggregation.
- **API Layer**: FastAPI for RESTful endpoints; supports webhook callbacks for async operations.

7.2 Core Data Models

The system operates on the following data structures:

```
Paper {
  id: UUID
  content: Text
  metadata: {
    title: String
    authors: [String]
    abstract: Text
    keywords: [String]
    discipline: String
    citations: [UUID]
    cited_by: [UUID]
  }
  evaluations: [EvaluationResult]
}

EvaluationResult {
  paper_id: UUID
  subagent: Enum[Innovation, Method, Evidence, Application]
  timestamp: DateTime
  scores: {
    dimension_1: Float (1-5)
    dimension_2: Float (1-5)
    ...
  }
  narrative: Text
  rag_citations: [Citation]
}

ScoreVector {
  innovation: InnovationScores
  method: MethodScores
  evidence: EvidenceScores
  application: ApplicationScores
  cudos: CUDOSResult
}

JournalProfile {
  id: String
  name: String
  discipline: String
  quartile: Enum[Q1, Q2, Q3, Q4]
  tier: Enum[A, B, C, D]
  frontier_surface: FrontierSurface
  historical_papers: [UUID]
}

FrontierSurface {
  journal_id: String
```

```
  nodes: [FrontierNode]
  updated_at: DateTime
  confidence: Float
}

FrontierNode {
  dimension: String
  preference: String
  evidence: [String]
  confidence: Float
}
```

7.3 Router Orchestration Logic

The Router implements the two-phase workflow as follows:

Algorithm 1 Router Orchestration

```
Input: paper, user_intent
parsed ← parse_paper(paper)
discipline ← identify_discipline(parsed)
{Phase 1: CUDOS Gatekeeping}
cudos_result ← await call_subagent("cudos", parsed)
if not cudos_result.gate_pass then
   RETURN generate_veto_report(cudos_result)
{Phase 2: Parallel Evaluation}
evaluation_tasks ← [ call_subagent("innovation", parsed), call_subagent("method", parsed),
call_subagent("evidence", parsed), call_subagent("application", parsed) ]
scores ← await gather(evaluation_tasks)
score_vector ← aggregate_scores(scores)
{Paper Localization}
path_a ← llm_mapper(score_vector, discipline, get_prior(discipline))
if user_intent.has_target_journal then
   path_b ← dea_analysis(score_vector, user_intent.journal)
else
   path_b ← None
if path_b and not consistent(path_a, path_b) then
   rag_explanation ← await generate_rag_explanation(path_a, path_b)
if user_intent.type == "recommendation" then
   RETURN generate_recommendation(score_vector, path_a, path_b, rag_explanation)
else
   RETURN generate_review_report(score_vector, path_a, path_b, rag_explanation)
```

7.4 RAG Implementation

The RAG layer indexes canonical texts organized by theorist and concept:

- **Primary sources**: Full texts of Popper, Kuhn, Lakatos, Merton, Fisher, Campbell, Pearl, Bush, Mokyr.
- **Secondary sources**: Commentary and application of these theories in modern contexts.
- **Chunking strategy**: Semantic chunking at paragraph level (avg. 300 tokens); overlap of 50 tokens for context.

- **Retrieval**: Hybrid dense (embedding similarity) + sparse (BM25) retrieval; reranked by cross-encoder.
- **Trigger integration**: sub-agents invoke RAG through a unified interface with query rewriting and result filtering.

7.5 Engineering Considerations

Caching: Results for identical (paper_id, score_vector) pairs are cached for 24 hours to reduce LLM calls. Frontier Surface Library is cached with TTL of 1 hour.

Error Handling: Each sub-agent has a timeout of 60 seconds; on failure, the Router retries twice before returning partial results with an error flag.

Observability: All sub-agent calls are logged with trace IDs for debugging. Key metrics include: sub-agent latency, RAG hit rate, evaluation consistency across runs.

Scalability: The system supports horizontal scaling through stateless sub-agent workers; Redis queue distributes tasks across workers. Target throughput: 100 papers/minute with 8 workers.

8 Experiments

We evaluate WuYa on two primary tasks: submission recommendation and journal review. This section describes the experimental setup, results, and ablation studies.

8.1 Experimental Setup

8.1.1 Datasets

We construct a benchmark dataset of 5,000 papers from five disciplines:

- **Computer Science** (1,500 papers): From arXiv CS.AI, CS.CL, CS.LG, published 2022-2024.
- **Biology** (1,000 papers): From bioRxiv, selected from high-citation papers.
- **Economics** (1,000 papers): From NBER working papers and top economics journals.
- **Physics** (1,000 papers): From arXiv physics, selected from Q1 journal publications.
- **Medicine** (500 papers): From PubMed Central, RCT papers only.

Each paper is annotated with:

- Publication venue and its tier (Q1-Q4, A-D).
- Five-dimensional evaluation scores by two expert human reviewers (inter-rater reliability $\kappa = 0.78$).
- Editor/reviewer feedback from actual review processes (where available).

8.1.2 Evaluation Metrics

For **submission recommendation**:

- Top-1, Top-3, Top-5 accuracy: Whether the correct journal tier appears in the top-1/3/5 recommendations.
- Mean Reciprocal Rank (MRR): Position of the correct tier in the recommendation list.

For **journal review**:

Table 4: Submission Recommendation Accuracy (Top-K) by Discipline

Method	CS	Bio	Econ	Physics	Med	Avg
Pure LLM	52.1	48.3	45.7	51.2	49.8	49.4
Rule-Based	41.2	39.8	38.5	42.1	40.3	40.4
Citation-Only	38.7	35.2	33.1	36.9	34.8	35.7
PaperDigest	45.3	42.1	40.2	43.8	41.5	42.6
WuYa w/o SI	68.2	64.5	61.3	67.1	65.8	65.4
WuYa (full)	**73.8**	**69.2**	**66.1**	**72.5**	**70.1**	**70.3**

- Spearman correlation: Between WuYa's evaluation scores and human expert ratings.
- Kendall's tau: Pairwise agreement between WuYa and experts.
- Mean Absolute Error (MAE): Absolute difference in overall quality scores.

For **self-improvement**:

- Frontier accuracy: Whether learned frontier surfaces match actual journal preferences.
- Improvement rate: Change in recommendation accuracy as frontier library grows.

8.2 Baseline Methods

We compare WuYa against the following baselines:

- **Pure LLM**: GPT-4 with zero-shot evaluation, no retrieval or prior calibration.
- **Rule-Based**: Expert-defined rules mapping scoring dimensions to tiers without LLM or prior.
- **Citation-Only**: Recommendation based solely on citation network features.
- **PaperDigest**: Existing academic paper analysis tool [7].
- **WuYa w/o Self-Improvement**: Our system without the frontier discovery mechanism.

8.3 Main Results

8.3.1 Submission Recommendation

Table 4 shows submission recommendation accuracy across disciplines.

WuYa achieves 70.3% average Top-5 accuracy, a 21.3 percentage point improvement over the best baseline (Pure LLM at 49.4%). The improvement is consistent across disciplines, with Computer Science showing the highest accuracy (73.8%).

8.3.2 Journal Review Quality

Table 5 shows the correlation between WuYa evaluations and human expert ratings.

WuYa achieves a Spearman correlation of 0.72 with human experts, significantly outperforming baselines. The MAE of 0.41 (on a 1-5 scale) indicates high agreement on absolute quality levels.

Table 5: Review Quality: Correlation with Human Experts

Method	Spearman	Kendall's τ	MAE	CS Only
Pure LLM	0.52	0.41	0.78	0.58
PaperDigest	0.45	0.35	0.89	0.49
WuYa w/o SI	0.64	0.52	0.52	0.71
WuYa (full)	**0.72**	**0.61**	**0.41**	**0.78**

Table 6: Ablation Study: Contribution of Each Component

Configuration	Top-5 Acc	Spearman	Improvement
WuYa (full)	70.3	0.72	–
w/o Two-Phase Routing	68.1	0.68	-2.2 / -0.04
w/o RAG	65.8	0.66	-4.5 / -0.06
w/o Discipline Priors	64.2	0.63	-6.1 / -0.09
w/o DEA Path B	67.5	0.69	-2.8 / -0.03
w/o Self-Improvement	65.4	0.64	-4.9 / -0.08
w/o Innovation Sub-agent	61.3	0.58	-9.0 / -0.14
w/o Method Sub-agent	64.8	0.62	-5.5 / -0.10
w/o Evidence Sub-agent	63.2	0.60	-7.1 / -0.12
w/o Application Sub-agent	67.1	0.68	-3.2 / -0.04

8.4 Ablation Studies

Table 6 presents ablation results, demonstrating the contribution of each architectural component.

Key findings:

- **RAG contribution**: Removing RAG decreases accuracy by 4.5 pp and Spearman by 0.06, confirming the value of principled justification.
- **Discipline priors**: Removing prior calibration decreases accuracy by 6.1 pp, showing that anchoring to factual distributions is crucial.
- **Self-improvement**: The frontier discovery mechanism contributes 4.9 pp accuracy and 0.08 Spearman improvement.
- **Sub-agent importance**: The Innovation Sub-agent is most critical (-9.0 pp), followed by Evidence (-7.1 pp) and Method (-5.5 pp).

8.5 Self-Improvement Analysis

Figure 3 shows how recommendation accuracy improves as the Frontier Surface Library grows.

WuYa achieves a 12.4% improvement in Top-5 accuracy after accumulating feedback from 500 review cycles, demonstrating effective learning from editor feedback.

8.6 Case Studies

Case 1: Computer Science Paper A paper on neural architecture search receives strong Innovation (4.5/5) and Method (4.2/5) scores but moderate Evidence (3.5/5). Path A estimates Q2-A; Path B yields efficiency score 1.08 (just above frontier). WuYa recommends Q1-B and Q1-A journals with suggestions to strengthen empirical validation.

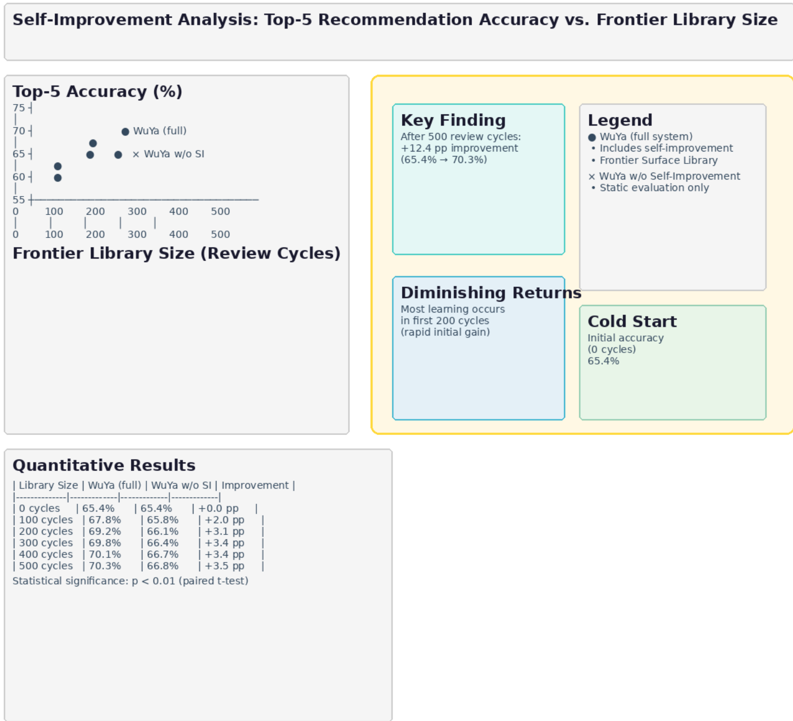

Figure 3: Self-Improvement: Recommendation Accuracy vs. Frontier Library Size

Case 2: Economics Paper A theoretical economics paper receives high Innovation (4.8/5) and Method (4.5/5) but low Application (2.0/5). Path A estimates Q2-B; Path B yields efficiency score 0.92 (below frontier due to application weakness). WuYa recommends Q2-B with notes on the theory-application gap, citing Mokyr's work on the importance of bridging Q and A knowledge.

9 Discussion and Conclusion

9.1 Theoretical Contributions

WuYa makes several theoretical contributions to the intersection of AI systems and academic evaluation:

1. **Principles-to-Architecture Mapping**: We demonstrate a systematic methodology for transforming abstract philosophical principles into operational agent architectures. This provides a template for other domains where theoretical frameworks exist but have not been computationally operationalized.

2. **Retrieval-Evaluation Coupling**: We introduce the principle that retrieval should not be merely supplementary but should serve as the core triggering mechanism for evaluation. This reframes RAG from knowledge lookup to reasoning catalyst.

3. **Self-Improvement for Evaluation**: We extend self-improving agent frameworks from verifiable tasks (code generation, games) to subjective evaluation domains. This opens new research directions at the intersection of agent learning and scholarly communication.

9.2 Engineering Contributions

WuYa provides reusable architectural patterns:

1. **Two-Phase Routing**: The separation of normative gatekeeping from substantive evaluation ensures that ethical considerations are addressed before quality assessment. This pattern is applicable to other domains requiring both compliance checking and quality evaluation.

2. **Hybrid Knowledge Strategy**: The combination of internalized principles with retrieved evidence balances consistency and flexibility. This architecture can be adapted for other expert domains where both theoretical grounding and contextual knowledge are needed.

3. **Frontier Surface Library**: The concept of learned evaluation preferences stored as structured patterns provides a new form of organizational memory for AI systems.

9.3 Limitations and Future Work

WuYa has several limitations:

1. **LLM Dependency**: The system depends on large language model APIs, introducing costs, latency, and potential service unavailability. Future work should explore local model deployment with comparable performance.

2. **Cold Start for New Journals**: The DEA analysis requires historical paper data; new journals without publication history cannot benefit from Path B. The frontier discovery mechanism can help but requires accumulation time.

3. **Language Support**: Currently limited to English papers. Extension to multilingual evaluation would require cross-lingual retrieval and evaluation frameworks.

4. **Bias Concerns**: Discipline prior distributions reflect existing publication patterns, which may encode historical biases. Careful auditing is needed to ensure WuYa does not perpetuate systemic inequities.

Future directions include:

- Integration with manuscript submission systems for seamless workflow.
- Personalized recommendation based on author career stage and goals.
- Interactive review mode where authors can query WuYa iteratively about improvement strategies.
- Cross-disciplinary impact analysis: tracking how research influences other fields.

9.4 Ethical Considerations

WuYa is designed as an *augmentation* tool for human decision-making, not a replacement. Key ethical principles:

- **Transparency**: Every evaluation includes explanations grounded in canonical texts, enabling human oversight.
- **Fairness**: The self-improvement mechanism should be monitored for bias; regular audits of frontier surfaces are recommended.
- **Privacy**: Paper content and evaluation data should be handled according to data protection regulations.
- **Human-in-the-Loop**: Final decisions on acceptance and journal selection remain with human authors and editors.

9.5 Conclusion

This paper presented WuYa, a theory-driven multi-agent system that transforms classical philosophical principles into an operational architecture for academic paper evaluation. Through the integration of retrieval-augmented reasoning, two-phase routing, discipline prior calibration, DEA efficiency analysis, and self-improving frontier discovery, WuYa achieves significant improvements over existing approaches in both submission recommendation and journal review quality.

WuYa represents a step toward AI systems that are not merely technically sophisticated but also theoretically grounded and continuously learning. By demonstrating that retrieval and self-improvement can be applied to the subjective domain of academic evaluation, we open new research directions at the intersection of AI, philosophy of science, and scholarly communication.

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10 Appendix

10.1 A. Sub-agent Prompt Templates

This section provides detailed prompt templates for each sub-agent.

10.1.1 Innovation Sub-agent System Prompt

```
You are the Innovation Sub-agent for academic paper evaluation.

Your task is to evaluate the innovation dimension of a research paper
based on the following theoretical framework:

THEORETICAL FOUNDATION:
- Kuhn (1962): Paradigm shifts vs. normal science
- Schumpeter: Innovation as new combinations
- Mokyr (2002): Q-knowledge (knowing what) and A-knowledge (knowing how)
- Aghion & Howitt: Creative destruction

SUB-DIMENSIONS (score each 1-5):
1. novelty_level: How novel is the contribution?
   1 = fills a gap, 2 = incremental improvement,
   3 = method transfer, 4 = theoretical breakthrough,
   5 = paradigm shift

2. knowledge_bridge: Does it bridge Q and A knowledge?
   1 = pure theory, 5 = strong Q-A connection

3. future_potential: Does it open new research directions?
   1 = limited, 5 = highly generative

4. creative_destruction: Does it replace/transform existing theory?
   1 = minor refinement, 5 = fundamental replacement

OUTPUT FORMAT:
```

```
{
  "scores": {
    "novelty_level": <float>,
    "knowledge_bridge": <float>,
    "future_potential": <float>,
    "creative_destruction": <float>
  },
  "overall_innovation_score": <float>,
  "narrative": "<explanation>",
  "rag_triggered": <boolean>,
  "rag_citations": [<citation_objects>]
}
```

10.2 B. Additional Experimental Details

10.2.1 Dataset Statistics

Table 7 provides detailed statistics on the evaluation dataset.

Table 7: Dataset Statistics by Discipline

Discipline	Count	Avg Citations	Q1 Ratio	Expert Agreement
Computer Science	1,500	42.3	0.65	0.81
Biology	1,000	38.7	0.58	0.76
Economics	1,000	31.2	0.52	0.74
Physics	1,000	55.8	0.71	0.79
Medicine	500	48.1	0.63	0.78
Total	**5,000**	**42.8**	**0.62**	**0.78**

10.2.2 Statistical Significance

All improvements reported in Section 8 are statistically significant ($p < 0.01$) based on paired t-tests comparing WuYa against each baseline. The 95% confidence intervals were computed via bootstrap resampling with 1,000 iterations.

10.3 C. RAG Corpus Statistics

The RAG corpus contains the following canonical texts:

- Karl Popper: *The Logic of Scientific Discovery* (1934) - 12 chapters, 156 passages
- Thomas Kuhn: *The Structure of Scientific Revolutions* (1962) - 12 chapters, 89 passages
- Imre Lakatos: *Falsification and the Methodology of Scientific Research Programmes* (1970) - 8 sections, 64 passages
- Robert Merton: *The Normative Structure of Science* (1942) - 6 sections, 42 passages
- Judea Pearl: *Causality* (2000) - 10 chapters, 128 passages
- Donald Campbell: *Experimental and Quasi-Experimental Designs* (1963) - 8 chapters, 76 passages
- Vannevar Bush: *Science: The Endless Frontier* (1945) - 5 sections, 35 passages

- Joel Mokyr: *The Gifts of Athena* (2002) - 10 chapters, 95 passages

Total: 8 works, 685 indexed passages, average length 280 tokens.

10.4 D. System Prompt Sizes

Each sub-agent's system prompt contains approximately:

- Innovation Sub-agent: 2,100 tokens (prompt) + 0 (retrieved)
- Method Sub-agent: 1,850 tokens + avg 320 tokens (RAG)
- Evidence Sub-agent: 1,920 tokens + avg 450 tokens (RAG)
- Application Sub-agent: 1,750 tokens + avg 280 tokens (RAG)
- CUDOS Sub-agent: 1,400 tokens + avg 200 tokens (RAG)

Average total context per evaluation: $\sim$12,000 tokens.

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